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ORIGINAL RESEARCH

Priority-Based Adaptive Traffic Signal Control System for Smart Cities

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Abstract
The traffic signal control system has paramount importance in managing the traffic congestion at road intersections. It allows smooth traffic flow through the road intersections. Keeping in view to increase the traffic flow through the intersections, this paper proposes an adaptive traffic signal which can be used as a prototype for smart cities. The proposed traffic signal system will act according to the context of the traffic status in the intersections. The phase selection is adaptive in nature in the proposed system and phase will be selected based on the lane priority. The lane priority is calculated based on traffic parameters: vehicle density and waiting time. Also, the green signal time of the system of each cycle is dynamic. The amount of green time a lane will get depends on the vehicle density of that lane. The proposed system has the ability to handle the emergency vehicle approaching the intersection by shifting to emergency mode from regular mode. The simulation result shows that the average percentage of traffic flow has been increased by 31.05% in the proposed adaptive traffic signal control system as compared with the fixed time traffic signal system. It also performs comparatively well in both the traffic situations: balanced vehicle density and unbalanced vehicle density across the roads in the intersection.

Keywords Adaptive traffic signal · Fixed time control traffic signal · Smart cities · Intelligent traffic management system

Introduction

Due to the advancement of information and communication technology (ICT), the concept of a smart city becomes a reality. The main aim of a smart city is to enhance the standard of living of the citizens as well as to increase the economic growth [1]. Smart mobility is one of the significant parts of the smart city initiative [2]. The number of vehicles is increasing every passing day due to socioeconomical growth and migration of people from rural to urban area [3]. This leads to high traffic congestion and imposes a huge load on the transportation infrastructure. Transport authority faces a problem in managing traffic congestion. Traffic congestion led to different consequences like: air pollution, monetary loss, loss of valuable time. According to a world bank study, India experiences $6 billion monetary loss in a year due to traffic congestion [4]. Traffic signal has a major role for maintaining the smooth traffic flow through the intersections. The intersections are the most critical part of any road segments. The traffic bottleneck in the transportation network generally happens in the road intersections in a city and it has been seen that, 20% to 50% time of the total travel time has been spent waiting at the intersections [5]. For that, many techniques have been introduced like building flyovers, expansion the road lane in the intersections. But those are not feasible solution due to financial constraints.

The traffic signal system can play a very important role here to solve the traffic bottleneck in the road intersection. Fixed time traffic signal control is the most widely used traffic control system, but it cannot solve the traffic bottleneck in the road intersection [6]. In this type, traffic signal control system has a predetermined fixed green signal and red signal time for each lane. This leads to a unnecessary waiting of the vehicles in the road intersections when the traffic flow

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is uneven on the different lanes. The traffic control system which is adaptive in nature, can solve this issue. The adaptive type of system is the one which act according to the traffic status of the intersection. Here, the green and red signal time is not predetermined and fixed, rather it is dynamic. In this regard, this paper proposes a priority-based adaptive traffic signal which will change its green and red signal time of each lane in the intersection depending on traffic status. The proposed system first sense the traffic scenario in the intersection through some sensors and based on that, the priority of each lane will be calculated. According to the lane priority, phase will be selected for each cycle. Generally, phase is selected in round-robin fashion. But here, the phase selection will be dynamic, not in a fixed manner. Then the green signal time of each phase will be determined based on the vehicle density of the corresponding lane for each cycle. The proposed system will operate on two modes: regular mode and emergency mode. In case of emergency vehicle(EV) approaching an intersection , the system will be move to emergency mode so that the EV does not need to wait in the intersection.

The rest of the paper is organized as follows. "Related Work" focuses on the related work associated with the design of a traffic signal control system. The proposed system is described in "The Proposed System". "Simulation and Result Analysis" presents the simulation and result analysis of the proposed adaptive traffic signal system. Finally, "Conclusion" concludes the paper.

Related Work

Managing the traffic is becoming extremely difficult job for the transport authority due to the vehicular growth. Traffic signal control system is one of the most important part of a traffic management system (ITMS) for maintaining smooth traffic flow through the intersections. A good traffic signal control system can manage the traffic congestion in a city. That is why designing good traffic signal control system becomes the paramount importance to the researchers.

Traffic signal control system are classified into three categories: fixed time, adaptive, and actuated control system [7–9]. The fixed time traffic signal control system is the old-fashioned technique where the green signal, red signal, and amber signal time duration are fixed and predetermined. Here the signals will cycle through constantly in sequential order. Hence, all roads are served in a constant fixed ordered and get equal green signal time. But this type of signal strategy is effective where traffic is consistent but it is less efficient when the traffic is dynamic and unbalanced in nature [10, 11].

Machine learning techniques have been used in [12–17] for designing the adaptive and actuated traffic control system. In this work, Neural Networks, Q-Learning, Deep Reinforcement learning, Neuro-fuzzy Inference system are adopted to maximize the traffic flow rate, minimization of average delay and total travel time. The fuzzy programming technique has been used in [18] to solve the problem of traffic signal timing optimization where various optimization objectives has been assigned different weight coefficients. The weight coefficients changes based on the status of the traffic flow. Then the multi-objective function was transferred to single objective function and then it is solved using a genetic algorithm. In [12], nature-inspired algorithm also been used to make intelligent traffic control system. They have used cuckoo search technique to fine-tune the parameters of intelligent controller implemented using an artificial neural network. Another work [19] has been found where nature-inspired algorithm has been used. Here, authors proposed an algorithm which is based on simulated annealing and a genetic algorithm for optimizing the traffic signal time. The meta heuristic-based method also been used for traffic signal optimization in [20–22]. Here, authors have used PSO technique to optimize the traffic signal. In another study [23], the authors treat the scheduling of traffic light as a model-based optimization problem and they have used discrete harmony search technique to solve the problem. In another work [24], authors formulated a linear programming problem to optimize the traffic light in the congested network. In a similar vein, researchers used mixed integer linear programming (MILP) technique to formulate the traffic signal timing optimization problem in [25–28].

In another study [29], authors have designed an embedded system for traffic signal control and it is based on the concept of priority. Here, the traffic density of the roads at the intersection are calculated and the road with highest vehicle density gets the priority over the other roads and gets the green signal for the traffic movement. In the above-mentioned works, like in [29], the starvation problem will arise. The roads with high vehicle density will serve the maximum number of green signal time and hence the vehicles on the roads with low vehicle density will face unnecessary waiting in the intersection. The fixed time control traffic signal has also drawbacks in case of inconsistent traffic in the intersection. The roads with high vehicle density will not get extra amount of green signal time here and it leads to a traffic congestion on that particular road. In [24], authors formulated a linear programming problem but it is very difficult to formulate the precise mathematical problem due to the complexity of the structure of the urban traffic system. To overcome the problem, this paper proposes an adaptive traffic signal control system which will change the green signal time with respect to the vehicle density and average waiting time. Also, the phase selection is dynamic in nature in the proposed system.

The Proposed System

The traffic signal control system plays a significant role in managing traffic congestion i.e., traffic bottleneck that occurs in the road intersection. But the traditional fixed time control traffic signal cannot reduce the traffic bottleneck in the road intersection. For that, an intelligent traffic signal control system needs to be designed which will adapt according to the traffic status at the intersection. This kind of traffic signal control system is one of the important parts of the intelligent transportation management system (ITMS) [3]. The main aim of this work is to design an adaptive traffic signal which will change its green time duration of each lane in the intersection based on traffic status. Also, the phase will be selected as per the traffic status, rather than predetermined order. Here, two traffic parameters have been chosen: traffic density and waiting time of the cars in each lane to infer the traffic status at a particular point of time. To capture the traffic parameters, wireless sensor network can be used. Also the proposed system is intelligent enough in managing the emergency vehicle (EV) approaching towards an intersection. Hence, the system has two mode of operations: regular mode and emergency mode.

Components of the System

The proposed system consists of mainly four components namely sensors, road-side unit (RSU), processing unit, and traffic light controller. The Fig. 1 represents the block diagram of the proposed system. The components are described as follows:

1. Sensors: Real time road traffic information (like traffic density, traffic flow) is the most important factor in controlling the traffic signal. Different types of in-road sensors like inductive loop detector, pneumatic tube, RFID etc. [30, 31] are deployed in the road segment to acquire the real time traffic information.

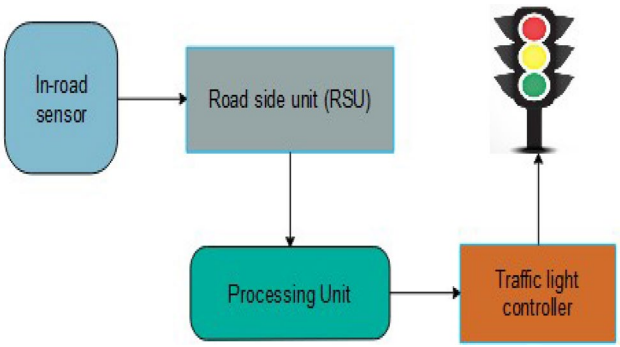


Fig. 1 Block diagram of the proposed system

2. Road side unit (RSU): A roadside unit is a dedicated short-range communications transceiver that is placed aside the road and it act as a getaway between the in-road sensors and processing unit. The sensed traffic information is sent to the processing unit and the processing unit sends the decision to the traffic light controller through RSU.
3. Processing unit: The processing unit is responsible for providing the adaptive nature of the traffic signal control system. It processes the real time in-road sensor data to determine the green signal time duration of each lane in the intersection. The phase is also selected based on the traffic status by the processing unit.
4. Traffic light controller: It is a device which is attached to the traffic light/signal. It basically controls the traffic light based on the instruction provided by the processing unit.

The proposed system can be implemented in a real world with the help of internet of things (IoT). The IoT based system architecture basically consist with the sensors, transceiver, processing unit and actuator so that things can sense and transmit the data to the processing unit and do the action based on the context of the environment. In the proposed system, traffic data are collected with the help of deployed sensors. These sensed traffic data are transmitted to the processing unit. The processing unit process the traffic data to determine the phase and green signal time duration of each lane in the intersection. This information then sent to the traffic light controller which will act as a actuator and based on that, it controls or adjust the traffic light/signal.

Working Principle

The proposed system will sense the traffic scenario in the intersection through some sensors and based on the traffic status, it will act. The system will decide two things initially listed as below, based on the context of the traffic environment: (A) Adaptive phase selection and (B) Determination of green signal duration.

A. Adaptive phase selection:The proposed system first selects the phase. A phase is sum of the green interval and clearance interval. The non-conflicting traffic movements are assigned in to each phase. In this work, both the 4-phase and 2-phase signals have been considered. The Figs. 2 and 3 shows the phase diagram of a four-legged intersection for 4-phase and 2-phase respectively. In 4-phase, the traffic can move in three directions (straight, left turn, and right turn) on each lane. Hence, movement of vehicles of Lane 1 in all three directions forms phase 1, movement of vehicles of Lane 2 in all three directions forms phase 2, movement of vehicles of Lane 3 in all

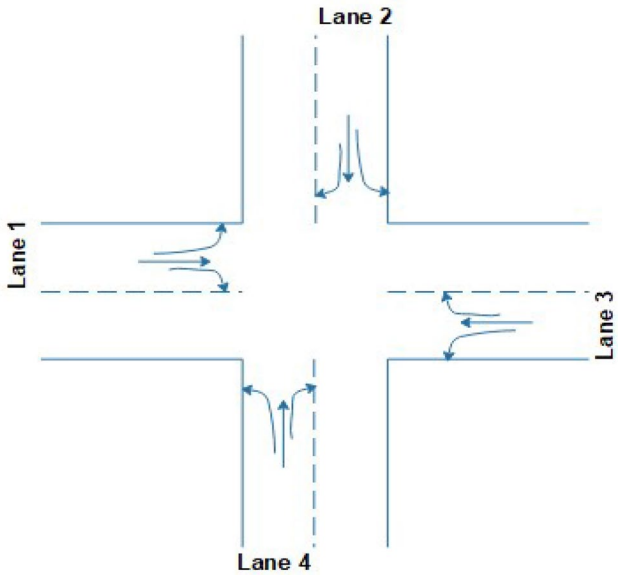


Fig. 2 Four-legged intersection with directed traffic flow (4-phase)

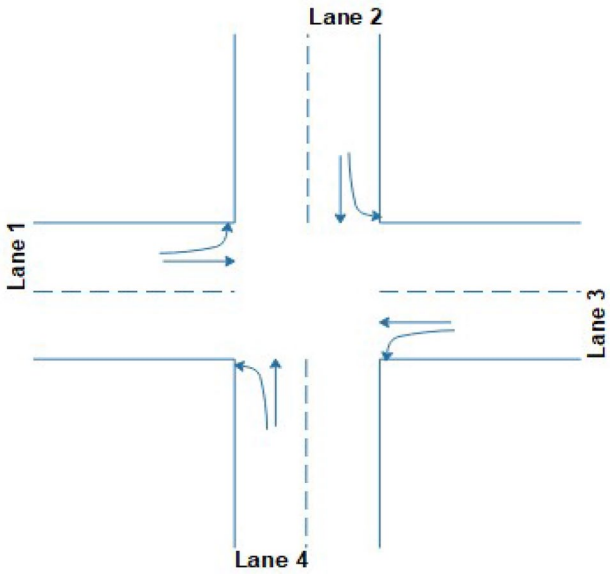


Fig. 3 Four-legged intersection with directed traffic flow (2-phase)

three directions forms phase 3, while the movement of vehicles of Lane 4 in all three directions forms phase 4. When one lane gets its green signal then all other 3 lane have red signal at the same point of time. Hence traffic of a particular lane can move in all three directions but the traffic of all other lanes have to wait at that time. So, no conflict will be there. In 2-phase, the traffic can move in two directions (straight and left turn) on each lane. Hence, the movement of vehicles of Lane 1 and Lane 3 forms phase 1 and movement of vehicles of Lane 2 and Lane 4 forms phase 2. Unlike traditional traffic system, the proposed traffic signal system has not any fixed order for phase selection. It will select the phase based on the priority. Hence, priority has to be calculated for each phase and it is denoted as $P(P_k)$. For that, first priority of each lane L_i has to be calculated and it is denoted as priority_L_i. The priority_L_i is the function of vehicle density vd_i and waiting time wt_i of the vehicles in the lane L_i. Here, the amount of time the vehicle belongs to a lane has to wait to get the next green signal indicates the waiting time. The wt_i can be calculated by subtracting the time of getting successive green signals of a lane. The combined value of vd_i and wt_i is mapped into the priority domain as shown in Fig. 4. It is represented as follows:

$$\text{priority_}L_i = f(wt_i, vd_i). \tag{1}$$

The priority of the lane L_i is proportional to both the vehicle density vd_i and the waiting time of the vehicles in the lane L_i, i.e., priority_$L_i \propto wt_i$, and priority_$L_i \propto vd_i$. Hence, the priority of the lane L_i is represented as:

$$f(wt_i, vd_i) = k.wt_i.vd_i. \tag{2}$$

While calculating the priority_L_i, the normalized value of vehicle density and normalized value of average waiting time of the vehicles are determined as follows:

$$\text{normalized}(wt_i) = \frac{wt_i - \min(wt_i)}{\max(wt_i) - \min(wt_i)}, \tag{3}$$

$$\text{normalized}(vd_i) = \frac{vd_i - \min(vd_i)}{\max(vd_i) - \min(vd_i)}. \tag{4}$$

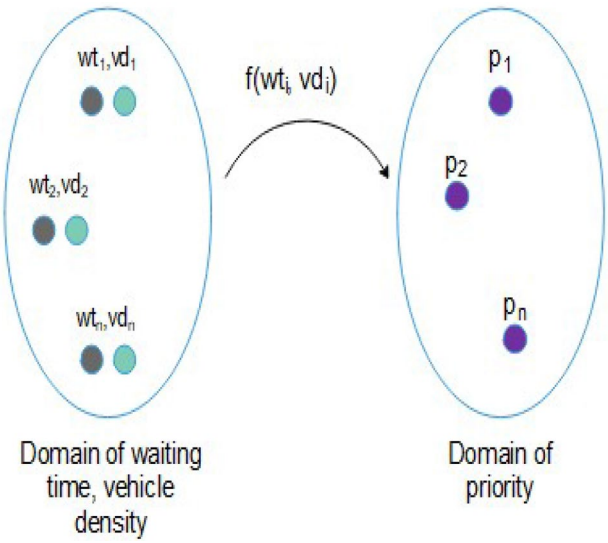


Fig. 4 Mapping of vd_i and wt_i with priority

Here, $\min(wt_i)$ is the minimum waiting time and $\max(wt_i)$ is the maximum waiting time of the vehicles in the intersection. $\min(vd_i)$ is the minimum vehicle density and $\max(vd_i)$ is the maximum vehicle density in the intersection. Hence, to determine the priority_L_i, the Eq. 2 has been rewritten and represented as follows:

$$f(wt_i, vd_i) = k.\text{normalized}(wt_i).\text{normalized}(vd_i). \quad (5)$$

While calculating the $P(P_k)$, this work considers two types of scenario in terms of phase formation. One is 4-phase and the other one is 2-phase signal. For 4-phase signal, the $P(P_k)$ is represented as:

$$P(P_k) = \text{priority_}L_i; \quad \text{where } k = i \text{ and } k = 1 \text{ to } 4. \quad (6)$$

For the 2-phase signal, the $P(P_k)$ is represented as:

$$P(P_k) = \max(\text{priority_}L_i, \text{priority_}L_j); \text{where } L_i, L_j \in P_k \text{ and } k = 1, 2. \quad (7)$$

B. Determination of green signal duration: The next step is to determine the green signal time of the selected phase in a signal cycle. A signal cycle is the one complete rotation of through all the traffic signal (i.e., green, amber and red). Here cycle length is fixed and, in this work, it is considered as 120 seconds. But in the proposed system, the green, red, and amber time duration in a cycle is not fixed or pre-determined. Rather, it also depends on the traffic status which is vehicle density vd_i of all the lanes in a junction. The green signal time duration of phase P_k in a particular cycle is $Gt(P_k)$ and calculated as follows:

$$Gt(P_k) = \frac{vd_i}{\text{total}(vd_i)}.\text{Available}(Gt). \quad (8)$$

Here, vd_i is the vehicle density at lane L_i during a particular cycle C_l. The $\text{total}(vd_i)$ is the sum of the vehicle density of all the lanes L_i in an intersection for the 4-phase signal and represented as follows:

$$\text{total}(vd_i) = \sum_{i=1}^{4} vd_i. \quad (9)$$

Whereas, the $total(vd_i)$ for the 2-phase signal is represented as follows:

$$\text{total}(vd_i) = \sum_{k=1}^{2} max(P_k(vd_i)). \quad (10)$$

Here, $\max(P_k(vd_i))$ is the maximum vehicle density among the lanes those belongs to phase P_k. The Available(Gt) is the actual available green time duration for a particular cycle C_l. This available green time duration has to be shared among all the four lanes in an intersection. The available green signal time Available(Gt) is presented as follows:

$$\text{Available}(Gt) = \text{CL} - \text{Amber_time}. \quad (11)$$

Here, CL is the cycle length and in this work, it is considered as 120 seconds. Amber_time is the total time duration of amber signal time of all the lanes and it is considered as 10 seconds.

As mentioned earlier that the proposed system has the capability of handling emergency vehicles (EV) and that is why the system has two modes of operations: regular mode (RM) and emergency mode (EM). The system will be in RM if no EV is approaching in the intersection and initially the system starts in RM. The proposed traffic signal system will be entered into EM from RM the moment it identifies an emergency vehicle approaches in the road intersection. The assumption is that EV must be equipped with communication unit so that the proposed system can detect the position of EV in the road network. If the EV is detected in a particular lane L_j of a road junction, then the system will first check whether the lane has the green signal or not. If it has, then the system will operate in regular mode. But if that lane has a red signal, then the proposed system will be shifted to emergency mode from regular mode. In this mode of operation, the system first identifies the lane L_i which has the green signal in this time. Then the system will assign amber and then red signal to this lane. And subsequently, the lane L_j which has the EV, will be assigned green signal for a specific time et. During et, the EV supposed to be pass through the intersection smoothly without waiting for the signal and once et time is over, immediately this lane will get the red signal. After that, the system will go back to the lane L_i which was the green signal and compensate the green signal time loss to lane L_i. For that, the et time will be added to the L_i as a green signal time. Subsequently, the green signal time for L_j will adjusted by $Gt(P_i) - et$. But if the EV is detected in a particular lane L_j and the lane has a green signal that moment then system will work in a regular mode. Algorithm 1 represents the working procedure of the proposed system.

Algorithm 1 Adaptive Traffic Signal Control

procedure ADAPTIVE TRAFFIC SIGNAL CONTROL
Sense the traffic parameters (vehicle density vd_i, average waiting time wt_i)
Calculate normalized score: $normalized(wt_i)$ and $normalized(vd_i)$
Determine $priority_L_i$ of each lane
Determine $P(P_k)$ of each phase
$Phase_order[]$:=ascending order of $P(P_k)$
Determine the $Gt(P_k)$ of each lane.
    if EV not detected **then**
        for $i \leftarrow 1$ to n **do**
            Assign $Gt(P_k)$ to $Phase_order[k]$
            Assign 2.5sec amber time to $Phase_order[k]$
            Assign $CL - (Gt(P_k) + ambertime)$ to $Phase_order[k]$
        end for
    else
        Detect EV approaching lane L_j
        Detect lanes L_i with green signal and red signal
        if L_j is red signal **then**
            Assign 2.5sec amber time to L_i
            Deduct green signal time by $Gt(P_k)$–et for L_j
            Add green signal time et for L_i
        else
            goto line no. 7
        end if
    end if
end procedure

Simulation and Result Analysis

In this section, the simulation of the proposed adaptive traffic signal system has been presented and the performance of the system is compared with the fixed-time signal. Python has been used to simulate the proposed system.

Simulation of the Proposed System

For the simulation, this work considers four-legged intersection as shown in Fig. 1. The simulation scenario has been presented in Table 1 and it will be used for 4-phase and 2-phase signal.

Table 1 shows that at time stamp $t1$, four lanes have varying vehicle density (*VD*) and average waiting time (*WT*). As per the proposed technique, the first step is the phase selection based on the priority $P(P_k)$. The Table 1 also shows the priority $_L_i$ of all the four lanes in the intersection at time stamp $t1$.

For the 4-phase signal, as the lane 2 (L2) has the priority 0.50, hence the priority of Phase 2 is 0.50 and it will be selected i.e., lane 2 will get the green time and vehicles of this lane can through the intersection in all three directions (i.e., straight, left turn, and right turn). In that time the other lanes in the intersection will have red signal.

Then the green signal time duration for the select phase $Gt(P_k)$ has been determined by the proposed system. In timestamp $t1$ the no. of vehicles in Lane 1 (L1) is 20, Lane 2 (L2) is 35, Lane 3 (L3) is 30, and Lane 4 (L4) is 20. Hence, the green signal time duration of L2 will be as follows as per Eq. 8:

Table 1 Simulation scenario

Lane ID	Time stamp: t1		
	VD (no. ofcars)	WT (in s)	Priority $_L_i$
L1	20	60	0.00
L2	35	70	0.50
L3	30	70	0.33
L4	20	70	0.00

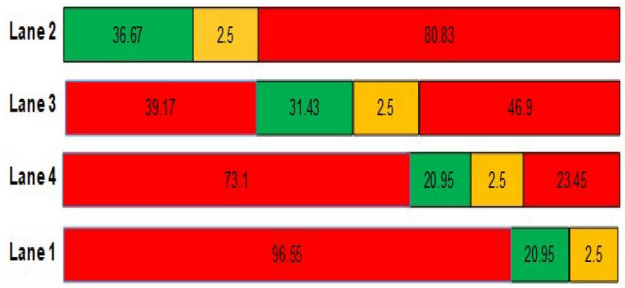


Fig. 5 Phase timing diagram of 4-phase signal



Fig. 6 Phase timing diagram of 2-phase signal

$$Gt(L_2) = 35/105 * 110 = 36.67\,\text{s}.$$

Then at time stamp $t1$, the second heights priority is 0.33 and hence Phase 3 will be selected i.e., the lane 3 (L3) will get the green time and vehicles of this lane can through the intersection in all three directions. Phase 1 and Phase 4 has the same priority as shown in Table 1. In that case, the phase will be selected which have the maximum waiting time in the intersection. So, Phase 4 will be selected next and the last phase is Phase 1 in this cycle. The green signal time duration of lane 3 (L3), lane 4 (L4), lane 1 (L1) will be as follows:

$$Gt(L_3) = 30/105 * 110 = 31.43\,\text{s}$$
$$Gt(L_4) = 20/105 * 110 = 20.95\,\text{s}$$
$$Gt(L_1) = 20/105 * 110 = 20.95\,\text{s}.$$

Hence, phase timing diagram of 4-phase signal at time stamp $t1$ will be as follows (Fig. 5).

For the 2-phase signal, phase 2 will be selected and hence the vehicles of lane 2 and 4 can move in two directions: straight and left turn. After that phase 1 will gets its turn. Now the green signal time of lane 2 and 4 will be:

$$Gt(L_2) = Gt(L_4) = 35/65 * 110 = 59.23\,\text{s}.$$

Then the green signal time of lane 1 and 3 will be:

$$Gt(L_1) = Gt(L_3) = 30/65 * 110 = 50.77\,\text{s}.$$

Hence, phase timing diagram of 2-phase signal at time stamp $t1$ will be as follows (Fig. 6).

Performance Metrics

To analyze the performance of the proposed system, this paper considers the following metrics and compares the performance of it with the fixed time traffic signal system:

- Percentage of traffic flow of each lane: It is defined as the average percentage of the traffic of a particular lane flow through the intersection at each time-stamp. It is represented as follows:

$$\%\ \text{of traffic flow} = \frac{\text{traffic flow}}{\text{vehicle density}} * 100. \tag{12}$$

Here,

$$\text{traffic flow} = \frac{Gt(P(P_k))}{h}. \tag{13}$$

h is in seconds and it represents the green time required in seconds by each vehicle to cross the intersection. Here, value of h is considered as 1.5 seconds. The more the Average percentage of traffic flow is, the better the system will be.

- Time gap between two successive green signals of a lane: It is defined as the amount of time that a lane has to wait to get a green signal from present cycle to next cycle.

Performance Analysis

To evaluate the performance of the proposed adaptive traffic signal control system, this work compares its results of it with the fixed time traffic signal control system based on the above-mentioned performance metrics. Any adaptive traffic signal system may suffer from starvation problem. Starvation means, a phase $P(P_k)$ has to wait a loner amount of time than usual to get the next green signal. That is why, the proposed system is compared with another adaptive traffic signal system [29] in this context. To conduct the simulation,

Table 2 Scenario 1

Timestamp	Vehicle density			
	Lane 1	Lane 2	Lane 3	Lane 4
t1	20	35	30	20
t2	32	53	35	41
t3	39	42	38	39
t4	36	52	44	42
t5	42	58	42	52
t6	28	63	33	53
t7	29	69	38	48

Table 3 Scenario 2

Timestamp	Vehicle density			
	Lane 1	Lane 2	Lane 3	Lane 4
t1	10	35	11	28
t2	22	42	12	29
t3	13	39	9	36
t4	13	44	8	42
t5	11	48	9	33
t6	10	51	9	38
t7	9	39	11	31

Python has been used. The simulation has been done on a four-legged intersection as shown in Fig. 2. Two scenarios, namely balanced and unbalanced vehicle density, have been considered for the simulation and these are presented in Tables 2 and 3 respectively.

Both the scenarios have four lanes in the intersection point. In this work, balanced vehicle density means all the lanes have almost the same number of vehicles or no huge difference between the number of vehicles across the lanes whereas unbalanced vehicle density means some lanes have high number of vehicles while other have few vehicles. In scenario 2, lane 1 and lane 3 has low vehicle density as compared to lane 2 and lane 4. If the vehicle density across the lane is unbalanced then the fixed time traffic signal system will fail to maintain the traffic flow due to its static nature. On the other hand, the adaptive system will able to maintain the traffic flow by adjusting phase and cycle length. The traffic signal system will be considered as a good one if it able to maintain smooth traffic flow (i.e., increase % of traffic flow) through intersection not only in a situation where traffic is balanced in the different lanes but also for the unbalanced traffic. These two different types of scenarios have been considered to analyze the performances of the proposed system. The proposed adaptive traffic signal control system and fixed time traffic signal control system have been implemented based on the scenario presented in Tables 2 and 3.

Average percentage of traffic flow of each lane has been calculated based on scenario 1 for both the traffic signal control system: adaptive and fixed time control. The Tables 4 and 5 shows the average percentage of traffic flow of each lane for the proposed adaptive traffic signal control system and the fixed time traffic signal control system respectively.

In this work, 27.5 s is considered as the green signal time duration for the fixed time traffic signal control system. This green signal time duration is fixed and predetermined. As the Available(Gt) = 110 s, this value is equally shared

Table 4 Percentage of traffic flow of each lane based on scenario 1 (proposed adaptive traffic signal control system)

Time stamp	Lane 1			Lane 2			Lane 3			Lane 4		
	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow
t1	20.95	13.97	69.83	36.67	24.45	69.85	31.43	20.95	69.84	20.95	13.97	69.83
t2	21.86	14.57	45.54	36.21	24.14	45.55	23.91	15.94	45.54	28.01	18.67	45.54
t3	27.15	18.1	46.41	29.24	19.49	46.41	26.46	17.64	46.42	27.15	18.1	46.41
t4	22.76	15.17	42.15	32.87	21.91	42.14	27.82	18.55	42.15	26.55	17.7	42.14
t5	23.94	15.96	38	33.05	22.03	37.99	23.37	15.58	37.09	29.64	19.76	38
t6	17.4	11.6	41.43	39.15	26.1	41.43	20.51	13.67	41.43	32.84	21.89	41.31
t7	17.34	11.56	39.86	41.25	27.5	39.86	22.71	15.14	39.84	28.7	19.13	39.86

Table 5 Percentage of traffic flow of each lane based on scenario 1 (fixed time traffic signal control system)

Time stamp	Lane 1			Lane 2			Lane 3			Lane 4		
	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow
t1	27.5	18.33	91.67	27.5	18.33	52.38	27.5	18.33	61.11	27.5	18.33	91.67
t2	27.5	18.33	57.29	27.5	18.33	34.59	27.5	18.33	52.38	27.5	18.33	44.72
t3	27.5	18.33	47.01	27.5	18.33	43.65	27.5	18.33	48.25	27.5	18.33	47.01
t4	27.5	18.33	50.93	27.5	18.33	35.26	27.5	18.33	41.67	27.5	18.33	43.65
t5	27.5	18.33	43.65	27.5	18.33	31.61	27.5	18.33	43.65	27.5	18.33	35.26
t6	27.5	18.33	65.48	27.5	18.33	29.1	27.5	18.33	55.56	27.5	18.33	34.59
t7	27.5	18.33	63.22	27.5	18.33	26.57	27.5	18.33	48.25	27.5	18.33	38.19

among the four phases. Each phase will get equal amount of green signal time (27.5 seconds) the fixed time traffic signal control system. From Table 4, it is seen that the average percentage of traffic flow of each lane in the proposed adaptive traffic signal control system is almost balanced but the average percentage of traffic flow of each lane in the fixed time traffic signal control system is not the balanced one as shown in Table 5. Though the % of traffic flow for Lane 1, Lane 3, and Lane 4 is high in fixed time traffic signal control system as compared to the proposed adaptive traffic signal control system, but in Lane 2, the % of traffic flow is very low in fixed time traffic signal control system. This happens because, in the fixed time traffic signal control system the green signal time, red signal time and amber signal time are predetermined and fixed. It does not depend upon the traffic status (i.e., vehicle density, waiting time) of the lanes. Even though in Lane 2, the vehicle density is high as shown in Table 2, still the green signal time will not increase to allow the more vehicle to pass through the intersection in the fixed time traffic signal control system. Hence this leads to the traffic congestion in Lane 2.

The performance of both the system is also compared for the scenario 2 as shown in Table 3. The Table 6 and 7 shows the average percentage of traffic flow of each lane in the proposed adaptive traffic signal control system and fixed time traffic signal control system respectively. The % of traffic flow in each lane in the proposed adaptive traffic signal control system is quite balanced. But in the fixed time traffic signal control system, the % of traffic flow in Lane 2 and Lane 4 is low as compared to the proposed system as shown in Fig. 7 and 8. This happens because even though these two lanes have higher vehicle density, still the green signal time duration will not adapt (i.e., increase) to allow the more vehicles to pass through. Hence, this may lead to traffic congestion in these two lanes. The simulation result shows that average % of traffic flow has been increased by 31.05% in the proposed traffic signal system as compared with the fixed time traffic control system.

It may happen that to serve green signal time more to the high vehicle density lane other low vehicle density lane may have to wait longer. This phenomenon is known as starvation. But the proposed system does not suffer from this kind of problem. To evaluate this, "Time gap between two successive green signals of a lane" performance metric has been used.

Figure 9 shows the time gap between two successive green signals in Lane 3 with respect to scenario 2. As the Lane 3 has the lowest vehicle density as compared to other lanes in scenario 2, that is why Lane 3 has been consider for the evaluation of the proposed system. From Fig. 9 it

Table 6 Percentage of traffic flow of each lane based on scenario 2 (proposed adaptive traffic signal control system)

Time stamp	Lane 1			Lane 2			Lane 3			Lane 4		
	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow
t1	13.09	8.73	87.27	45.83	30.55	87.3	14.4	9.6	87.27	36.66	24.44	87.29
t2	23.05	15.37	69.85	44	29.33	69.84	12.57	8.38	69.83	30.38	20.25	69.84
t3	14.74	9.83	75.59	44.23	29.49	75.61	10.21	6.81	75.63	40.82	27.21	75.59
t4	13.36	8.91	68.51	45.23	30.15	68.53	8.22	5.48	68.5	43.18	28.79	68.54
t5	11.98	7.99	72.61	52.28	34.85	72.61	9.8	6.53	72.59	35.94	23.96	72.61
t6	10.19	6.79	67.93	51.94	34.63	67.9	9.17	6.11	67.93	38.7	25.8	67.89
t7	11	7.33	81.48	47.66	31.77	81.47	13.44	8.96	81.45	37.89	25.26	81.48

Table 7 Percentage of traffic flow of each lane based on scenario 2 (fixed time traffic signal control system)

Time stamp	Lane 1			Lane 2			Lane 3			Lane 4		
	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow	Green signal	Traffic flow	% of traffic flow
t1	27.5	18.33	100	27.5	18.33	52.38	27.5	18.33	100	27.5	18.33	65.48
t2	27.5	18.33	100	27.5	18.33	43.68	27.5	18.33	100	27.5	18.33	63.22
t3	27.5	18.33	100	27.5	18.33	47.01	27.5	18.33	100	27.5	18.33	50.93
t4	27.5	18.33	100	27.5	18.33	41.67	27.5	18.33	100	27.5	18.33	43.65
t5	27.5	18.33	100	27.5	18.33	38.19	27.5	18.33	100	27.5	18.33	55.56
t6	27.5	18.33	100	27.5	18.33	35.95	27.5	18.33	100	27.5	18.33	48.25
t7	27.5	18.33	100	27.5	18.33	47.01	27.5	18.33	100	27.5	18.33	59.14

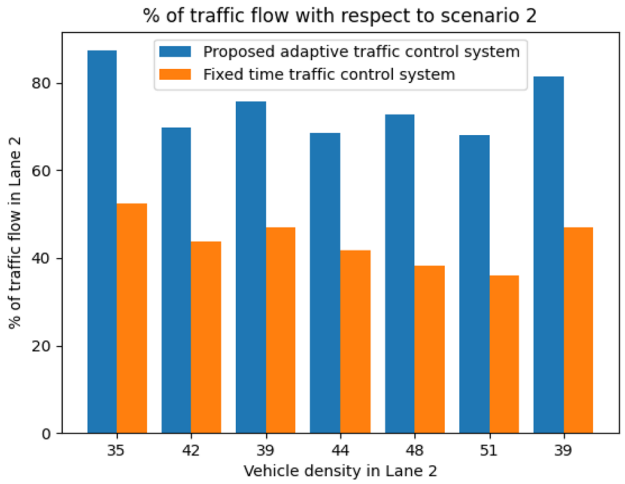


Fig. 7 % of traffic flow in Lane 2 with respect to scenario 2

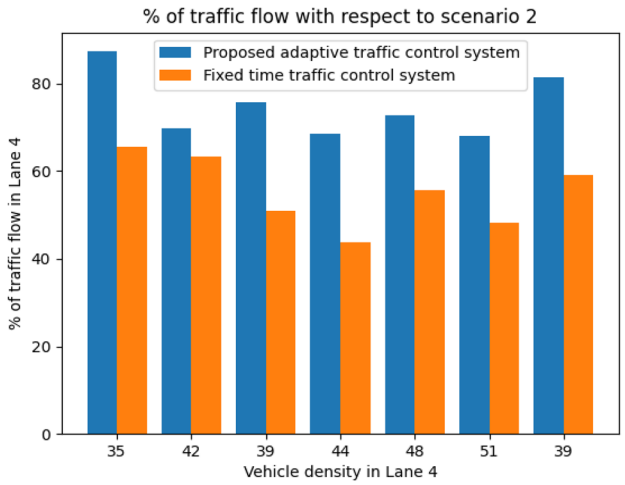


Fig. 8 % of traffic flow in Lane 4 with respect to scenario 2

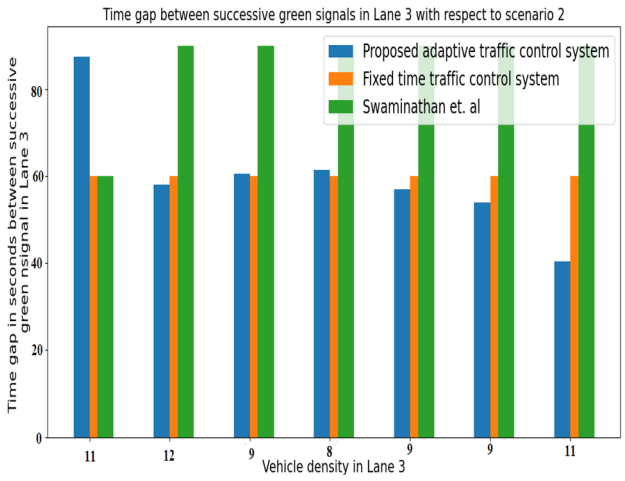


Fig. 9 Time gap between two successive green signals in Lane 3 with respect to scenario 2

is clearly seen that, the proposed system outperforms the system proposed in [29] in the context of starvation. In [29], authors consider only vehicle density to determine the phase section (i.e., Lane). The lane with heights vehicle density will always gets the priority and that is why the lanes with lower vehicle density will be starved for getting the next green signal time. And for the Fixed time control system, time gap between two successive green signals is always be equal. Hence, starvation problem is not an issue for fixed time control system.

Another point is that the traffic flow on one intersection will not hampers or effects the traffic flow in the next adjacent intersection with the use of proposed traffic signal control system. Because, whatever the traffic density will be in the next intersection, the proposed system will act as per the traffic status of that intersection.

Conclusion

This paper proposes an priority-based adaptive traffic signal control system which will act according to the context of the traffic status i.e., vehicle density and waiting time in the intersection. In the proposed system the phase selection is not in round robin fashion, rather it is in adaptive in nature. In addition, the duration of green signal time and red signal time is not fixed and predetermined. The duration of green signal time will also be adjusted depending upon the vehicle density of all the lanes in the intersection. Also, the proposed system has the ability to handle the emergency vehicles in the intersection. From the simulation results it is clearly seen that the proposed adaptive traffic signal control system performs comparatively well than the fixed time traffic signal control system in both the traffic situations: balanced vehicle density and unbalanced vehicle density across the lanes in the intersection. The % of traffic flow of a lane through the intersection is satisfactory even if it has high vehicle density. But this is not the case in the fixed-time traffic signal control system. Apart from that, the lane with low vehicle density is also not starved for getting the next green signal time. Hence, the proposed system can be used as a prototype model in the smart city to maintain the smooth traffic flow in the road intersection.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Mondal MA, Rehena Z. Intelligent traffic congestion classification system using artificial neural network. In: Companion Proceedings of the 2019 world wide web conference. Association for Computing Machinery, New York, NY, USA. 2019. pp. 110–6. https://doi.org/10.1145/3308560.3317053.
2. Monzon A. Smart cities concept and challenges: Bases for the assessment of smart city projects. In: 2015 international conference on smart cities and green ICT systems (SMARTGREENS). 2015. pp. 1–11.
3. Mondal MA, Rehena Z. An IoT-based congestion control framework for intelligent traffic management system. In: Chiplunkar NN, Fukao T, editors. Advances in artificial intelligence and data engineering. Singapore: Springer; 2021. p. 1287–97.
4. Rawal T. Intelligent transportation system in India—a review. J Dev Manag Commun. 2015;2:299.
5. Tarnoff PJ, Ordonez J. Signal timing practices and procedures-state of the practice. In: Transportation Research Board. 2005.
6. Bandra J. Traffic highway capacity design-traffic signal design. In: Traffic signal design. 2002.
7. Zhao D, Dai Y, Zhang Z. Computational intelligence in urban traffic signal control: a survey. IEEE Trans Syst Man Cybern Part C (Appl Rev). 2012;42(4):485–94. https://doi.org/10.1109/TSMCC.2011.2161577.
8. Papageorgiou M, Diakaki C, Dinopoulou V, Kotsialos A, Wang Y. Review of road traffic control strategies. Proc IEEE. 2003;91(12):2043–67. https://doi.org/10.1109/JPROC.2003.819610.
9. Ribeiro IM, de Lurdes de Oliveira Simões M. The fully actuated traffic control problem solved by global optimization and complementarity. Eng Optim. 2016;48(2):199–212. https://doi.org/10.1080/0305215X.2014.995644.
10. Webster FV. Traffic signal setting. Road Research Laboratory. 1958. pp. 1–44.
11. Miller AJ. Settings for fixed-cycle traffic signals. J Oper Res Soc. 1963;14(4):373–86. https://doi.org/10.1057/jors.1963.61.
12. Araghi S, Khosravi A, Creighton D. Intelligent cuckoo search optimized traffic signal controllers for multi-intersection network. Expert Syst. Appl. 2015. https://doi.org/10.1016/j.eswa.2015.01.063.
13. Jin J, Ma X, Kosonen I. An intelligent control system for traffic lights with simulation-based evaluation. Control Eng Pract. 2017;58:24–33. https://doi.org/10.1016/j.conengprac.2016.09.009.
14. Araghi S, Khosravi A, Creighton D, Nahavandi S. Influence of meta-heuristic optimization on the performance of adaptive interval type2-fuzzy traffic signal controllers. Expert Syst Appl. 2017;71(C):493–503. https://doi.org/10.1016/j.eswa.2016.10.066.
15. Miletić M, Kapusta B, Ivanjko E. Comparison of two approaches for preemptive traffic light control. In: 2018 international symposium ELMAR. 2018. pp. 57–62. https://doi.org/10.23919/ELMAR.2018.8534608.
16. Wei H, Zheng G, Yao H, Li Z. Intellilight: a reinforcement learning approach for intelligent traffic light control. In: Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery and data mining. KDD '18. Association for Computing Machinery, New York, NY, USA. 2018. pp. 2496–505. https://doi.org/10.1145/3219819.3220096.
17. Garg D, Chli M, Vogiatzis G. Deep reinforcement learning for autonomous traffic light control. In: 2018 3rd IEEE international conference on intelligent transportation engineering (ICITE). 2018. pp. 214–8. https://doi.org/10.1109/ICITE.2018.8492537.
18. Yu D, Tian X, Xing X, Gao S. Signal timing optimization based on fuzzy compromise programming for isolated signalized intersection. Math Probl Eng. 2016;2016:1–12. https://doi.org/10.1155/2016/1682394.
19. Li Z, Schonfeld P. Hybrid simulated annealing and genetic algorithm for optimizing arterial signal timings under oversaturated traffic conditions. J Adv Transp. 2014. https://doi.org/10.1002/atr.1274.
20. Gökçe M, Oner E, Ik G. Traffic signal optimization with particle swarm optimization for signalized roundabouts. Simulation. 2015;91:456–66. https://doi.org/10.1177/0037549715581473.
21. Dabiri S, Abbas M. Arterial traffic signal optimization using particle swarm optimization in an integrated vissim-matlab simulation environment. 2016. pp. 766–71. https://doi.org/10.1109/ITSC.2016.7795641.
22. Panovski D, Zaharia T. Simulation-based vehicular traffic lights optimization. In: 2016 12th international conference on signal-image technology internet-based systems (SITIS). 2016. pp. 258–65. https://doi.org/10.1109/SITIS.2016.49.
23. Gao K, Zhang Y, Sadollah A, Su R. Optimizing urban traffic light scheduling problem using harmony search with ensemble of local search. Appl Soft Comput. 2016. https://doi.org/10.1016/j.asoc.2016.07.029.
24. Eddelbuttel J, Cremer M. A new algorithm for optimal signal control in congested networks. J Adv Transp. 1994;28:275–97.
25. He Q, Kamineni R, Zhang Z. Traffic signal control with partial grade separation for oversaturated conditions. Transp Res Part C Emerg Technol. 2016;71:267–83. https://doi.org/10.1016/j.trc.2016.08.001.
26. Mehrabipour M, Hajbabaie A. A cell-based distributed-coordinated approach for network-level signal timing optimization. Comput Aided Civ Infrastruct Eng. 2017;32:599–616. https://doi.org/10.1111/mice.12272.
27. Köhler E, Strehler M. Traffic signal optimization: combining static and dynamic models. Transp Sci. 2018. https://doi.org/10.1287/trsc.2017.0760.
28. Yan H, He F, Lin X, Yu J, Li M, Wang Y. Network-level multiband signal coordination scheme based on vehicle trajectory data. Transp Res Part C Emerg Technol. 2019;107:266–86. https://doi.org/10.1016/j.trc.2019.08.014.
29. Swaminathan S, Venkatesan P. Embedded traffic control system using wireless ad hoc sensors. 2014. p. 225–7.
30. Lopes J, Bento J, Huang E, Antoniou C, Ben-Akiva M. Traffic and mobility data collection for real-time applications. In: 13th international IEEE conference on intelligent transportation systems. 2010. pp. 216–23. https://doi.org/10.1109/ITSC.2010.5625282.
31. Sumalee A, Ho HW. Smarter and more connected: future intelligent transportation system. IATSS Res. 2018;42(2):67–71. https://doi.org/10.1016/j.iatssr.2018.05.005.

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